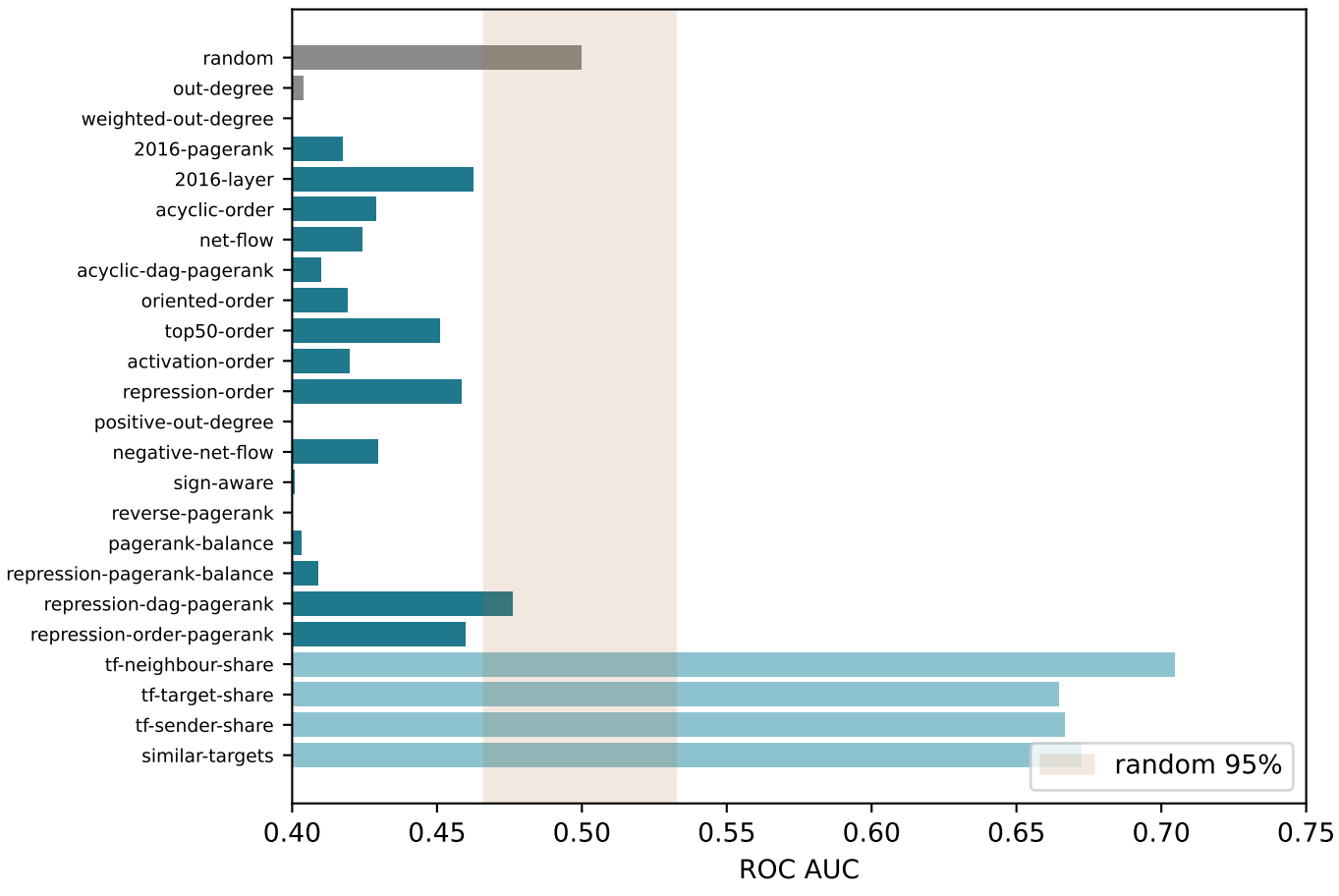
# Regulator ranking against reference 'dream5-tfs'

2026-11-08 · 4,511 genes ranked, 334 in tfs.txt · top k = 100

| Pipeline | Reference genes in top 100 | AUC | p top-k | p AUC | AUC vs best baseline (95%) | p vs baseline |
|---|---|---|---|---|---|---|
| random | 7.3 (3–12) | 0.500 (0.466–0.532) | | | | |
| out-degree | 6 | 0.404 | 0.749 | 1.000 | +0.000 (+0.000, +0.000) | |
| weighted-out-degree | 5 | 0.400 | 0.871 | 1.000 | -0.004 (-0.007, -0.001) | 0.026 |
| 2016-pagerank | 5 | 0.417 | 0.871 | 1.000 | +0.014 (-0.019, +0.050) | 0.424 |
| 2016-layer | 5 | 0.462 | 0.871 | 0.987 | +0.059 (+0.023, +0.092) | 0.002 |
| acyclic-order | 5 | 0.429 | 0.871 | 1.000 | +0.025 (+0.007, +0.045) | 0.010 |
| net-flow | 5 | 0.424 | 0.871 | 1.000 | +0.020 (-0.007, +0.050) | 0.144 |
| acyclic-dag-pagerank | 4 | 0.410 | 0.942 | 1.000 | +0.006 (-0.008, +0.022) | 0.424 |
| oriented-order | 3 | 0.419 | 0.981 | 1.000 | +0.015 (-0.020, +0.051) | 0.396 |
| top50-order | 9 | 0.451 | 0.301 | 0.997 | +0.047 (+0.016, +0.076) | 0.004 |
| activation-order | 3 | 0.420 | 0.981 | 1.000 | +0.016 (-0.006, +0.037) | 0.132 |
| repression-order | 9 | 0.458 | 0.301 | 0.992 | +0.054 (+0.021, +0.090) | < 0.001 |
| positive-out-degree † | 4 | 0.399 | 0.942 | 1.000 | -0.005 (-0.013, +0.003) | 0.202 |
| negative-net-flow † | 8 | 0.430 | 0.462 | 1.000 | +0.026 (-0.003, +0.056) | 0.084 |
| sign-aware † | 9 | 0.401 | 0.301 | 1.000 | -0.003 (-0.018, +0.012) | 0.664 |
| reverse-pagerank † | 3 | 0.380 | 0.981 | 1.000 | -0.023 (-0.034, -0.012) | < 0.001 |
| pagerank-balance † | 4 | 0.403 | 0.942 | 1.000 | -0.001 (-0.029, +0.029) | 0.926 |
| repression-pagerank-balance † | 1 | 0.409 | 0.999 | 1.000 | +0.005 (-0.024, +0.035) | 0.712 |
| repression-dag-pagerank † | 4 | 0.476 | 0.942 | 0.912 | +0.072 (+0.042, +0.102) | < 0.001 |
| repression-order-pagerank † | 8 | 0.460 | 0.462 | 0.990 | +0.056 (+0.026, +0.084) | < 0.001 |
| tf-neighbour-share ‡ | 36 | 0.705 | < 0.001 | < 0.001 | +0.301 (+0.254, +0.345) | < 0.001 |
| tf-target-share ‡ | 20 | 0.664 | < 0.001 | < 0.001 | +0.261 (+0.214, +0.306) | < 0.001 |
| tf-sender-share ‡ | 28 | 0.667 | < 0.001 | < 0.001 | +0.263 (+0.217, +0.309) | < 0.001 |
| similar-targets ‡ | 24 | 0.672 | < 0.001 | < 0.001 | +0.268 (+0.221, +0.316) | < 0.001 |

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines.
AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data.
‡: uses known TFs: scored from other genes' labels in this reference (lighter bars).

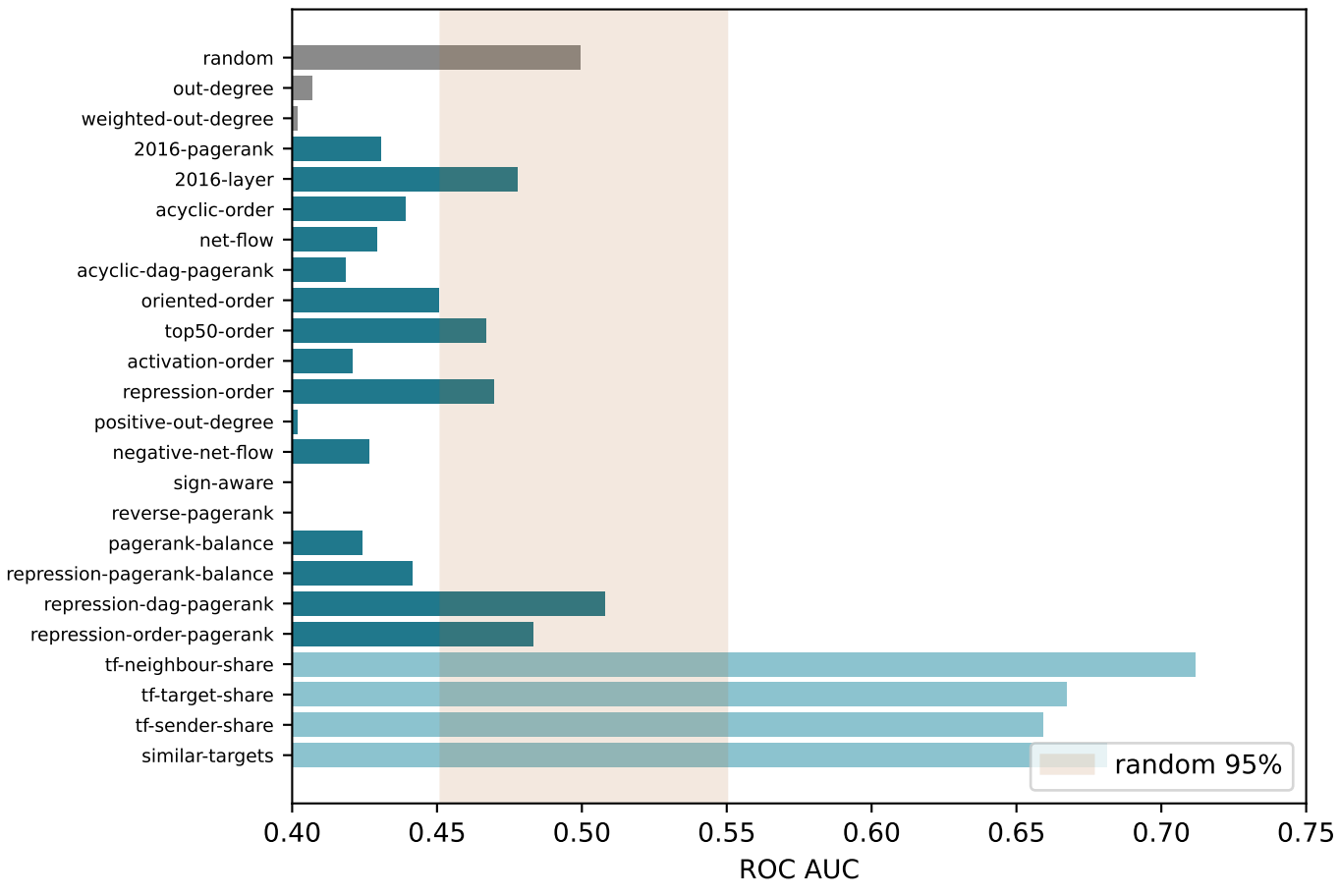


# Regulator ranking against reference 'true-regulators'

2026-11-08 · 4,511 genes ranked, 141 in regulators.txt · top k = 100

| Pipeline | Reference genes in top 100 | AUC | p top-k | p AUC | AUC vs best baseline (95%) | p vs baseline |
|---|---|---|---|---|---|---|
| random | 3.1 (0–7) | 0.499 (0.451–0.550) | | | | |
| out-degree | 2 | 0.407 | 0.830 | 1.000 | +0.000 (+0.000, +0.000) | |
| weighted-out-degree | 2 | 0.402 | 0.830 | 1.000 | -0.005 (-0.009, -0.002) | 0.008 |
| 2016-pagerank | 0 | 0.431 | 1.000 | 0.996 | +0.024 (-0.031, +0.075) | 0.428 |
| 2016-layer | 2 | 0.478 | 0.830 | 0.806 | +0.071 (+0.017, +0.126) | 0.018 |
| acyclic-order | 2 | 0.439 | 0.830 | 0.993 | +0.032 (+0.000, +0.063) | 0.048 |
| net-flow | 2 | 0.429 | 0.830 | 0.998 | +0.022 (-0.025, +0.066) | 0.352 |
| acyclic-dag-pagerank | 2 | 0.418 | 0.830 | 1.000 | +0.011 (-0.015, +0.036) | 0.392 |
| oriented-order | 2 | 0.451 | 0.830 | 0.977 | +0.044 (-0.013, +0.095) | 0.154 |
| top50-order | 6 | 0.467 | 0.080 | 0.908 | +0.060 (+0.012, +0.106) | 0.012 |
| activation-order | 2 | 0.421 | 0.830 | 1.000 | +0.014 (-0.020, +0.048) | 0.462 |
| repression-order | 2 | 0.470 | 0.830 | 0.886 | +0.063 (+0.009, +0.116) | 0.028 |
| positive-out-degree † | 0 | 0.402 | 1.000 | 1.000 | -0.005 (-0.018, +0.006) | 0.366 |
| negative-net-flow † | 3 | 0.427 | 0.604 | 0.998 | +0.020 (-0.027, +0.067) | 0.432 |
| sign-aware † | 3 | 0.397 | 0.604 | 1.000 | -0.011 (-0.033, +0.013) | 0.380 |
| reverse-pagerank † | 2 | 0.385 | 0.830 | 1.000 | -0.022 (-0.041, -0.002) | 0.034 |
| pagerank-balance † | 2 | 0.424 | 0.830 | 0.999 | +0.017 (-0.030, +0.063) | 0.508 |
| repression-pagerank-balance † | 1 | 0.441 | 0.958 | 0.987 | +0.034 (-0.016, +0.082) | 0.188 |
| repression-dag-pagerank † | 2 | 0.508 | 0.830 | 0.354 | +0.101 (+0.047, +0.151) | < 0.001 |
| repression-order-pagerank † | 3 | 0.483 | 0.604 | 0.732 | +0.076 (+0.026, +0.124) | 0.002 |
| tf-neighbour-share ‡ | 16 | 0.712 | < 0.001 | < 0.001 | +0.305 (+0.245, +0.364) | < 0.001 |
| tf-target-share ‡ | 11 | 0.667 | 0.001 | < 0.001 | +0.260 (+0.201, +0.319) | < 0.001 |
| tf-sender-share ‡ | 13 | 0.659 | < 0.001 | < 0.001 | +0.252 (+0.191, +0.307) | < 0.001 |
| similar-targets ‡ | 10 | 0.681 | 0.003 | < 0.001 | +0.274 (+0.215, +0.333) | < 0.001 |

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines.
AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data.
‡: uses known TFs: scored from other genes' labels in this reference (lighter bars).



# Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

## gold

| Graph | Pairs | Recall (×) | Positive × | Negative × | Sign (chance) |
|---|---|---|---|---|---|
| raw graph | 2,066 | 0.113 (10.58×) | 13.59× | 0.40× | |
| resolve_reciprocals → drop_sinks → prune_triangles | 2,066 | 0.012 (4.03×) | 12.66× | 0.46× | |
| resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering | 1,638 | 0.007 (3.51×) | 17.73× | 0.00× | |
| absolute | 2,066 | 0.113 (10.58×) | | | |
| absolute → minimum_feedback_order | 1,918 | 0.060 (7.99×) | | | |
| orient_reciprocals ratio=2.0 → absolute | 2,066 | 0.027 (4.44×) | | | |
| orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order | 2,066 | 0.027 (4.44×) | | | |
| top_outgoing k=50 → absolute | 2,066 | 0.104 (14.13×) | | | |
| top_outgoing k=50 → absolute → minimum_feedback_order | 2,011 | 0.065 (12.51×) | | | |
| keep_sign sign=1 → absolute | 2,031 | 0.114 (13.72×) | | | |
| keep_sign sign=1 → absolute → minimum_feedback_order | 2,022 | 0.056 (10.15×) | | | |
| keep_sign sign=-1 → absolute | 2,034 | 0.001 (0.39×) | | | |
| keep_sign sign=-1 → absolute → minimum_feedback_order | 1,977 | 0.001 (0.44×) | | | |
| keep_sign sign=-1 | 2,034 | 0.001 (0.39×) | | 0.39× | |

# Pipeline steps

### out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 296,953 edges (56,437 negative), largest strong component 4,294.

### weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 296,953 edges (56,437 negative), largest strong component 4,294.

### 2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 67,719 edges (51,333 negative), largest strong component 3,354.

### 2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 67,719 edges (51,333 negative), largest strong component 3,354; DAG keeps 37,759 edges.

### acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 296,953 edges (0 negative), largest strong component 4,294; DAG keeps 212,162 edges.

### net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 296,953 edges (0 negative), largest strong component 4,294.

### acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 296,953 edges (0 negative), largest strong component 4,294; DAG keeps 212,162 edges.

### oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 161,440 edges (0 negative), largest strong component 1,556; DAG keeps 161,056 edges.

### top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 167,372 edges (0 negative), largest strong component 4,287; DAG keeps 118,493 edges.

### activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 240,516 edges (0 negative), largest strong component 4,292; DAG keeps 166,567 edges.

### repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 56,437 edges (0 negative), largest strong component 2,699; DAG keeps 50,936 edges.

### positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 296,953 edges (56,437 negative), largest strong component 4,294.

### negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 296,953 edges (56,437 negative), largest strong component 4,294.

### sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 296,953 edges (56,437 negative), largest strong component 4,294.

### reverse-pagerank: Highest reverse PageRank (raw graph)